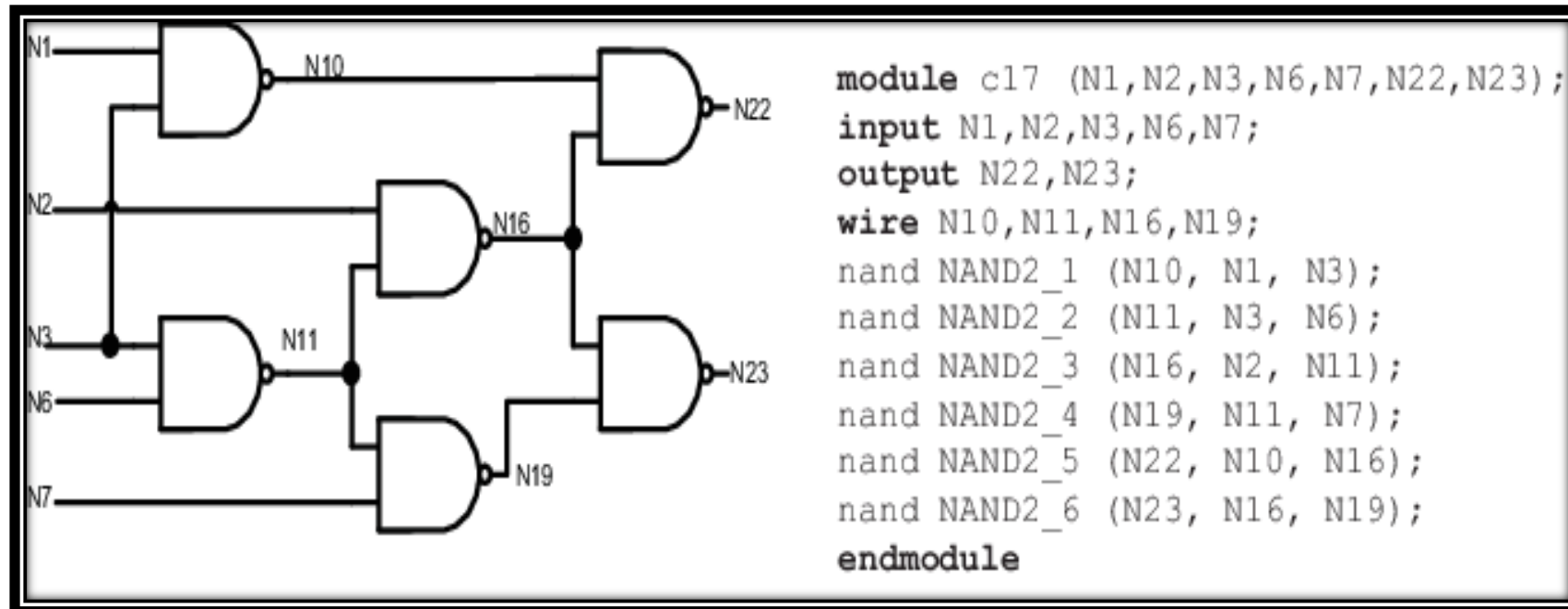




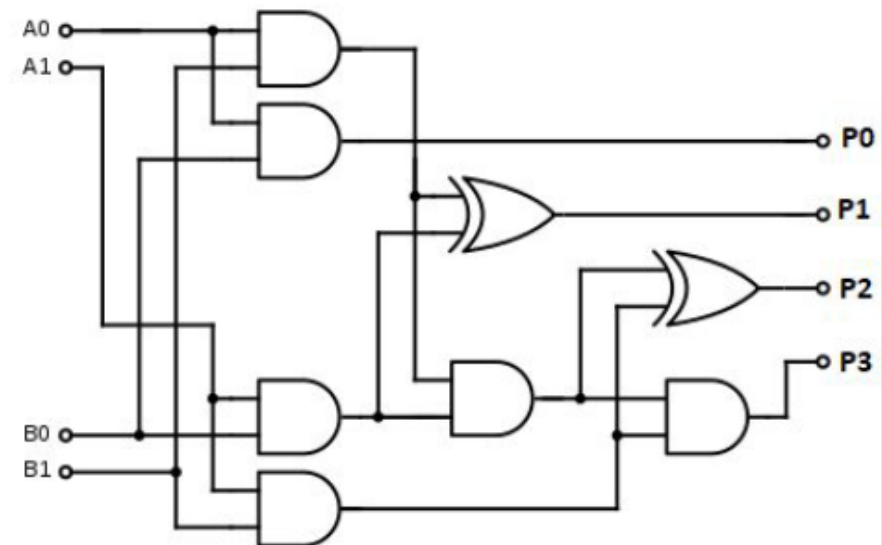
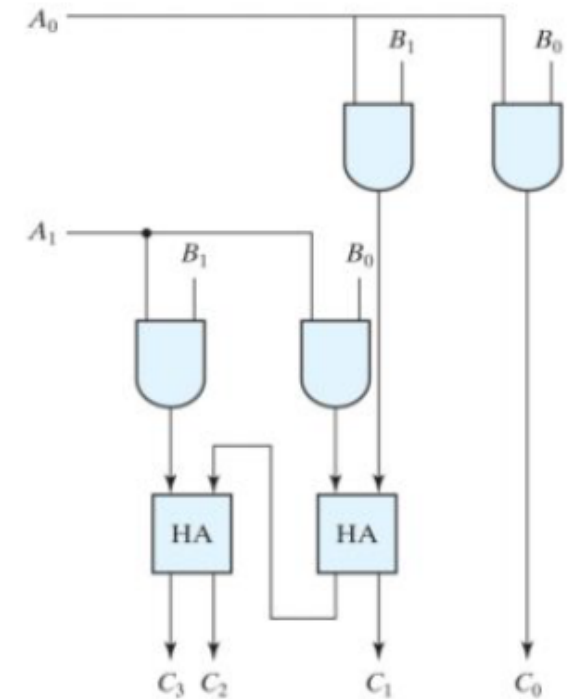
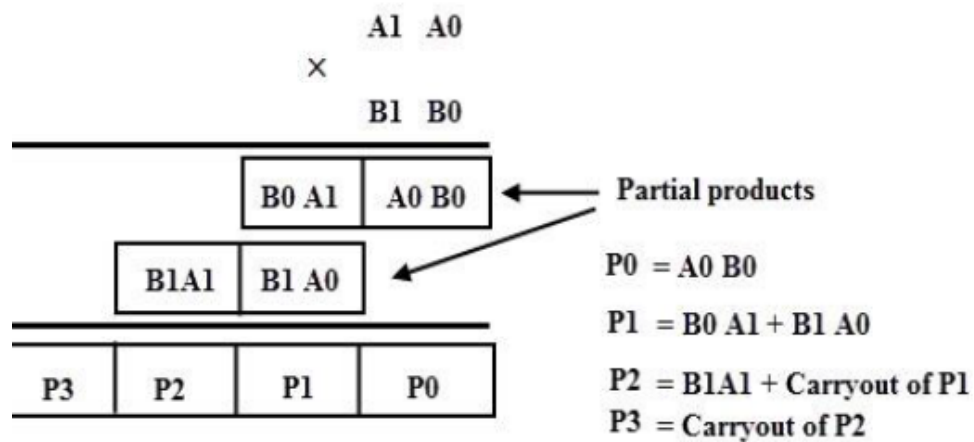
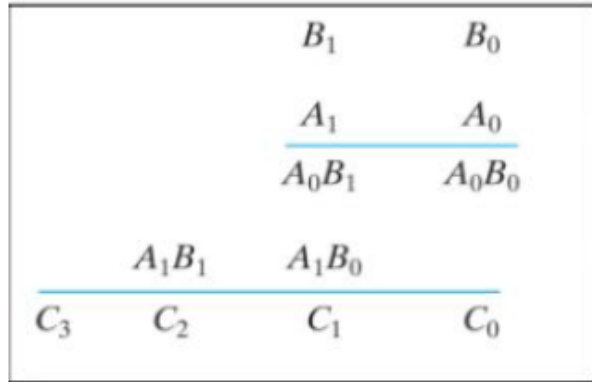
Digital Systems Design : BECE102L

Array Multiplier and Booth Algorithm



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Unsigned 2x2 Array Multiplier

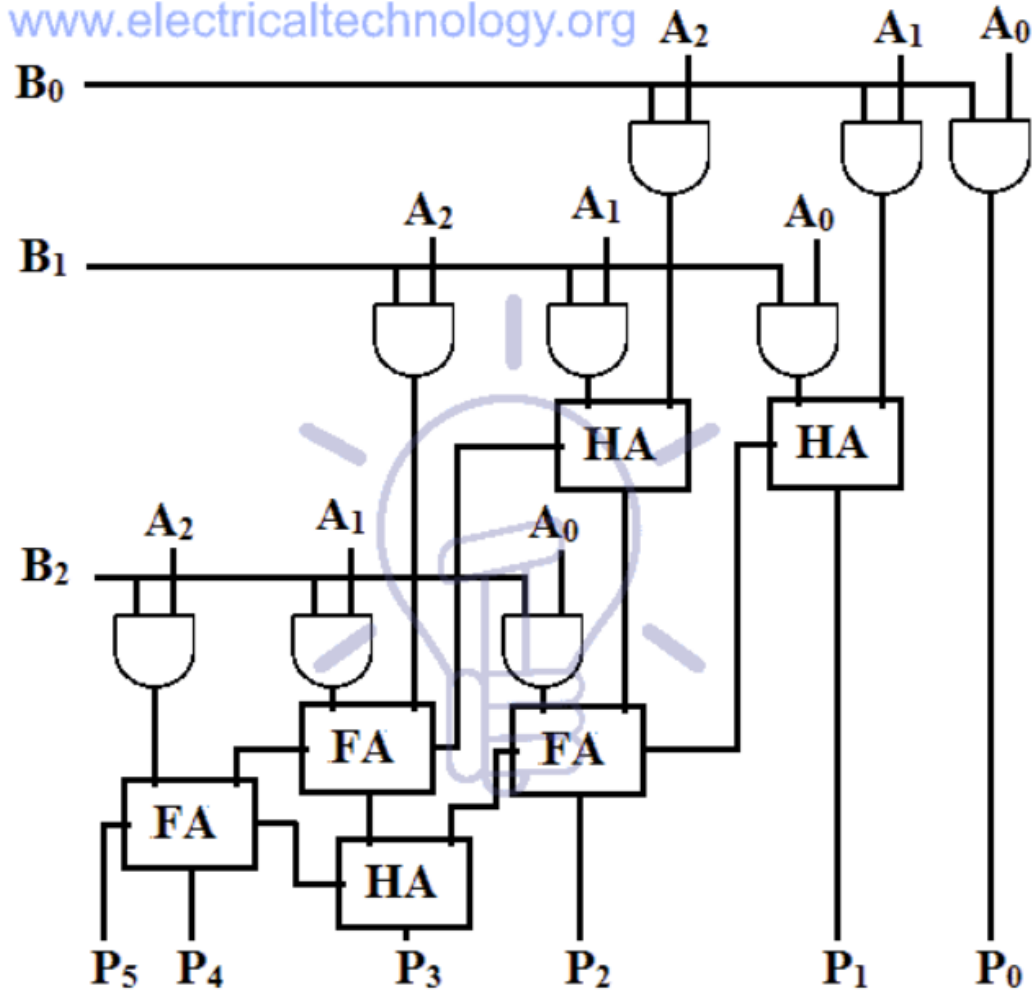


Unsigned 3*3 Multiplier

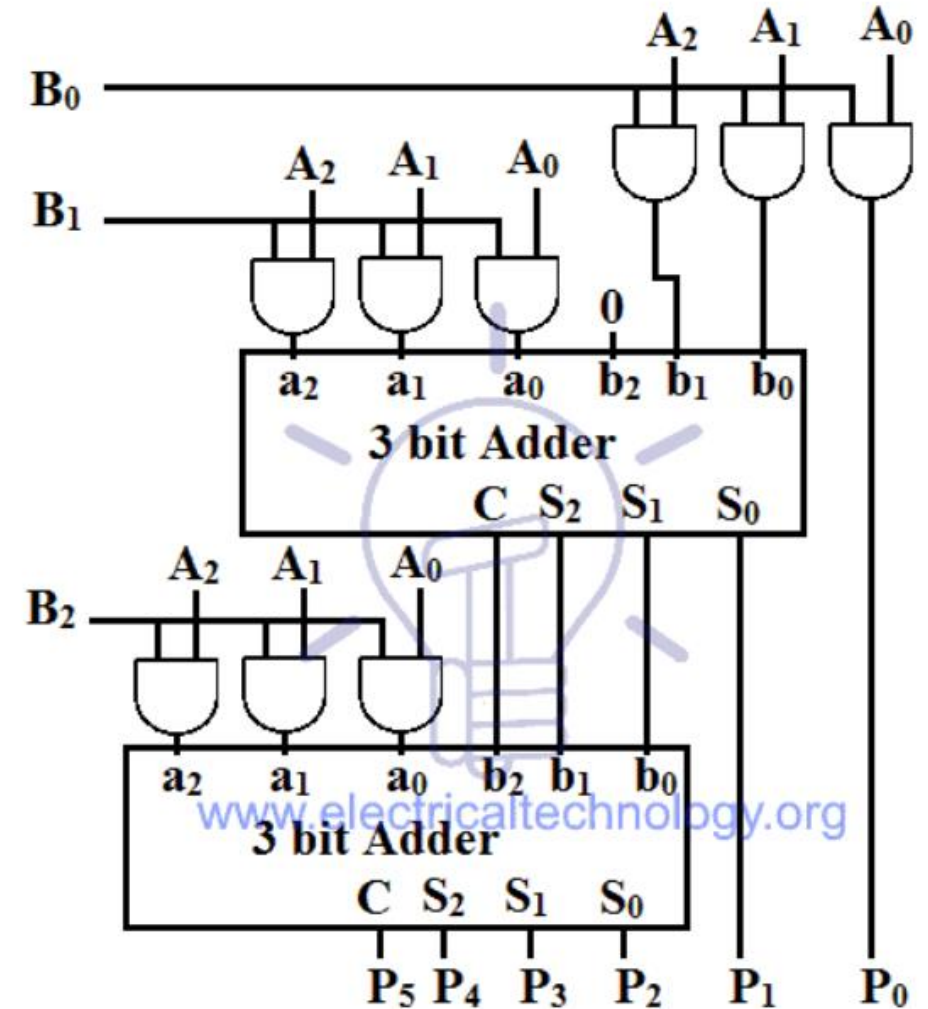
			A2	A1	A0	
			B2	B1	B0	
			A2B0	A1B0	A0B0	
		A2B1	A1B1	A0B1	X	
	A2B2	A1B2	A0B2	X	X	
A2B2+C+C	A2B1+A1B2	A1B1+C+	A0B1+A1B0	A0B0		
	+A2B2+C+C	A0B2+A2B0				

Unsigned 3*3 Multiplier

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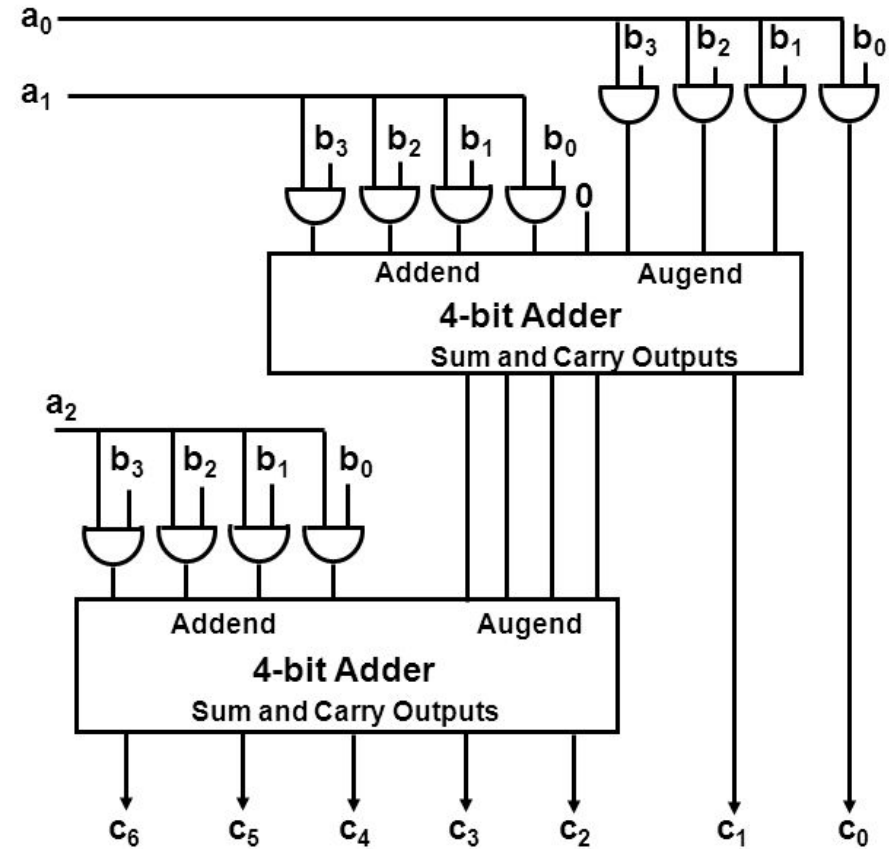
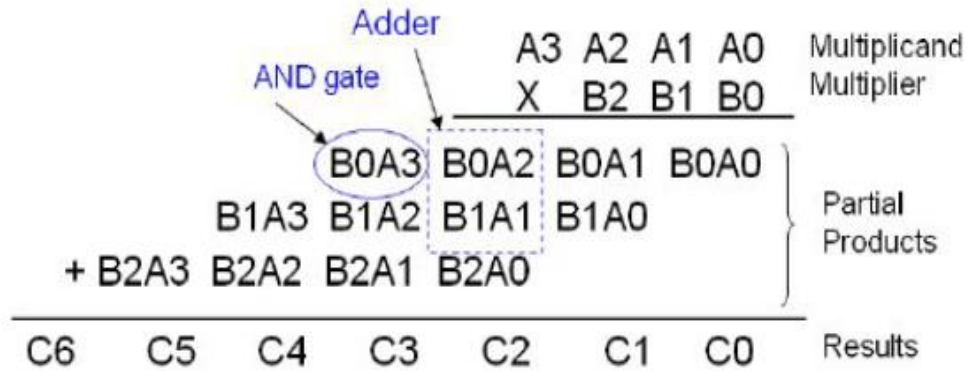


**Schematic of 3x3 Multiplier
Using Single-Bit Adder**

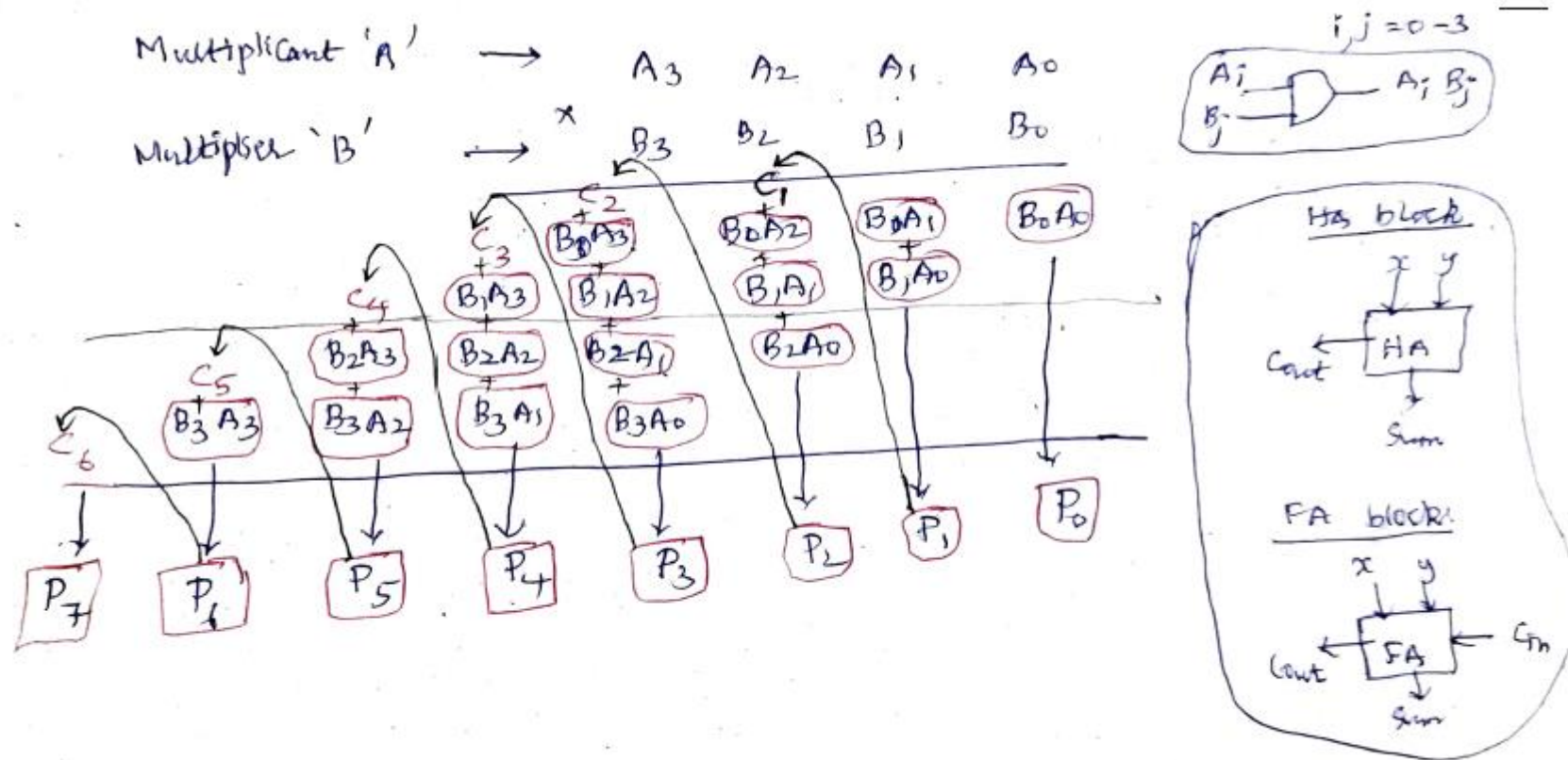


**Schematic of 3x3 Multiplier
Using 3-Bit Full Adder**

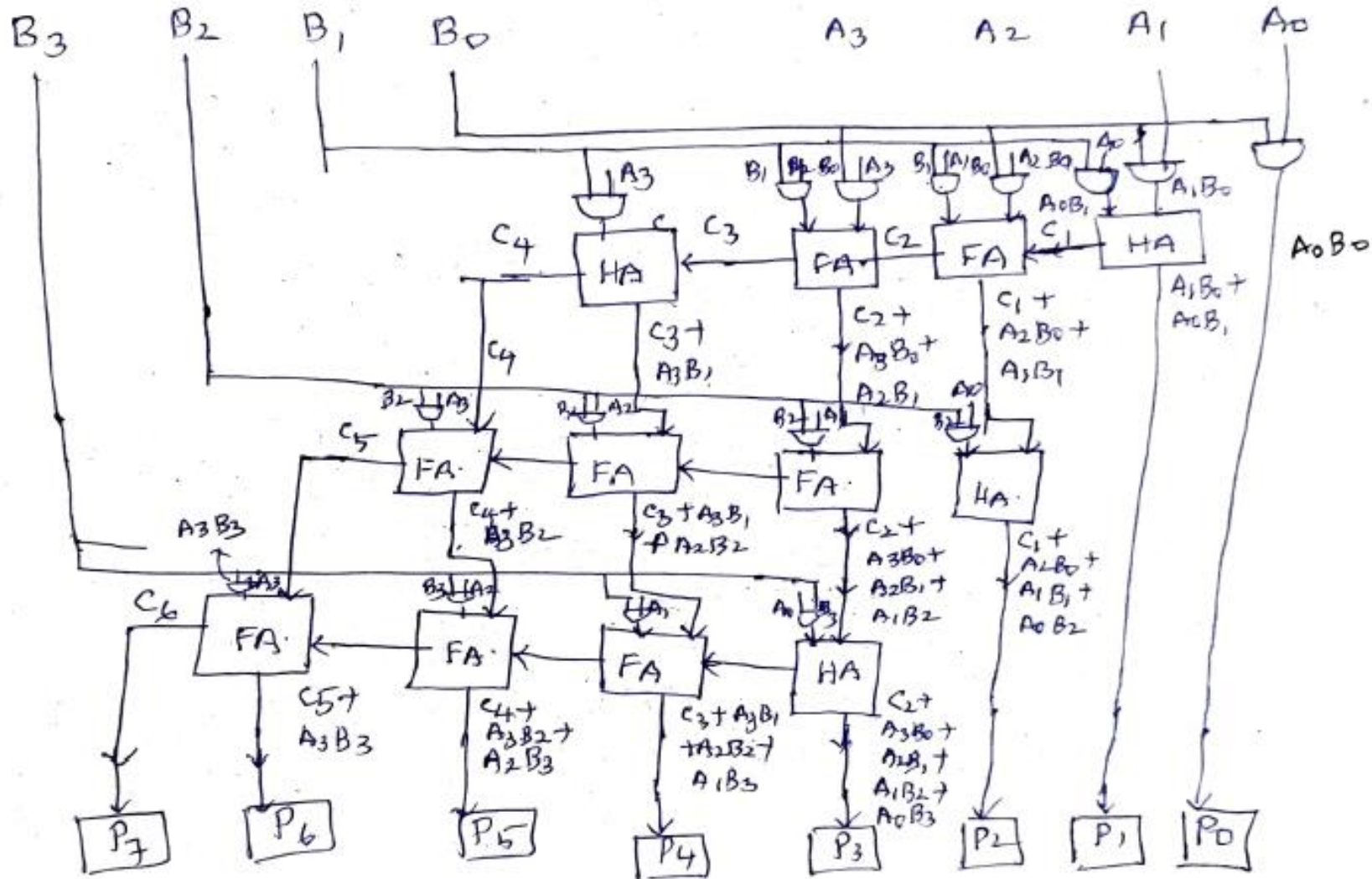
ARRAY MULTIPLIER 4-BIT X 3-BIT



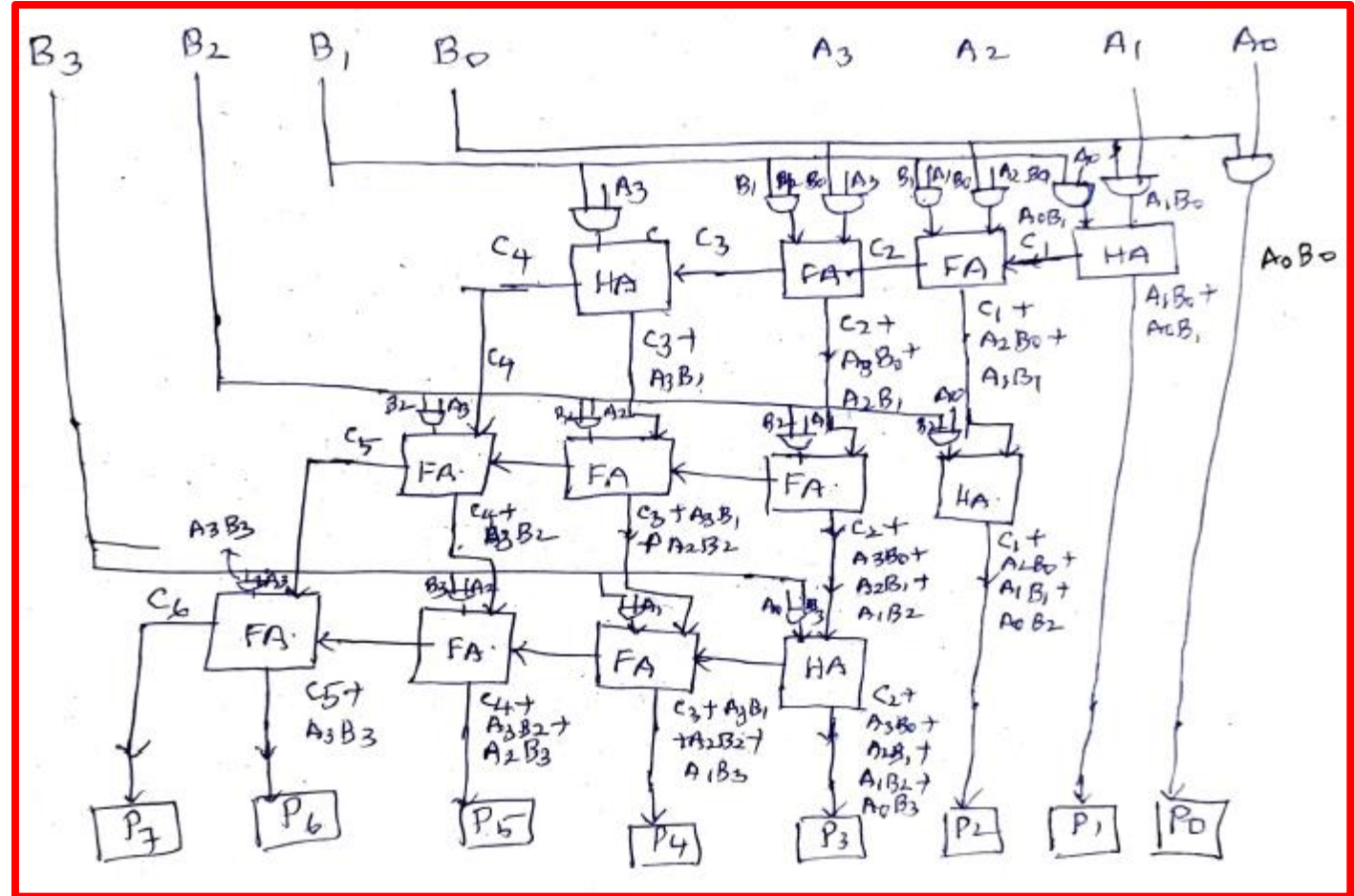
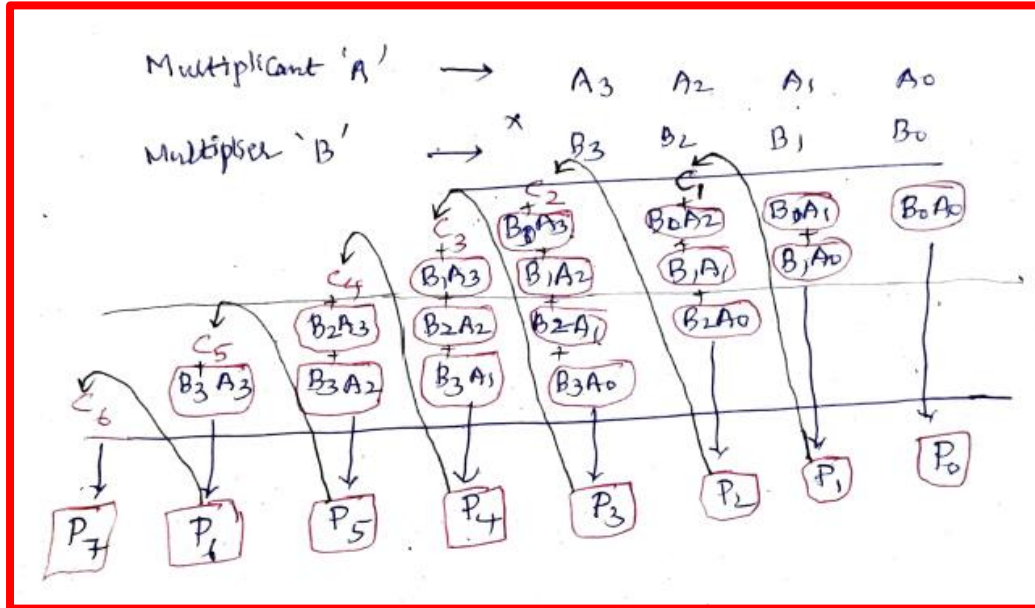
Unsigned 4*4 Multiplier



Unsigned 4*4 Multiplier



Unsigned 4*4 Multiplier



For 4x4 array multiplier requires,

16 - AND gates

4 - HAs

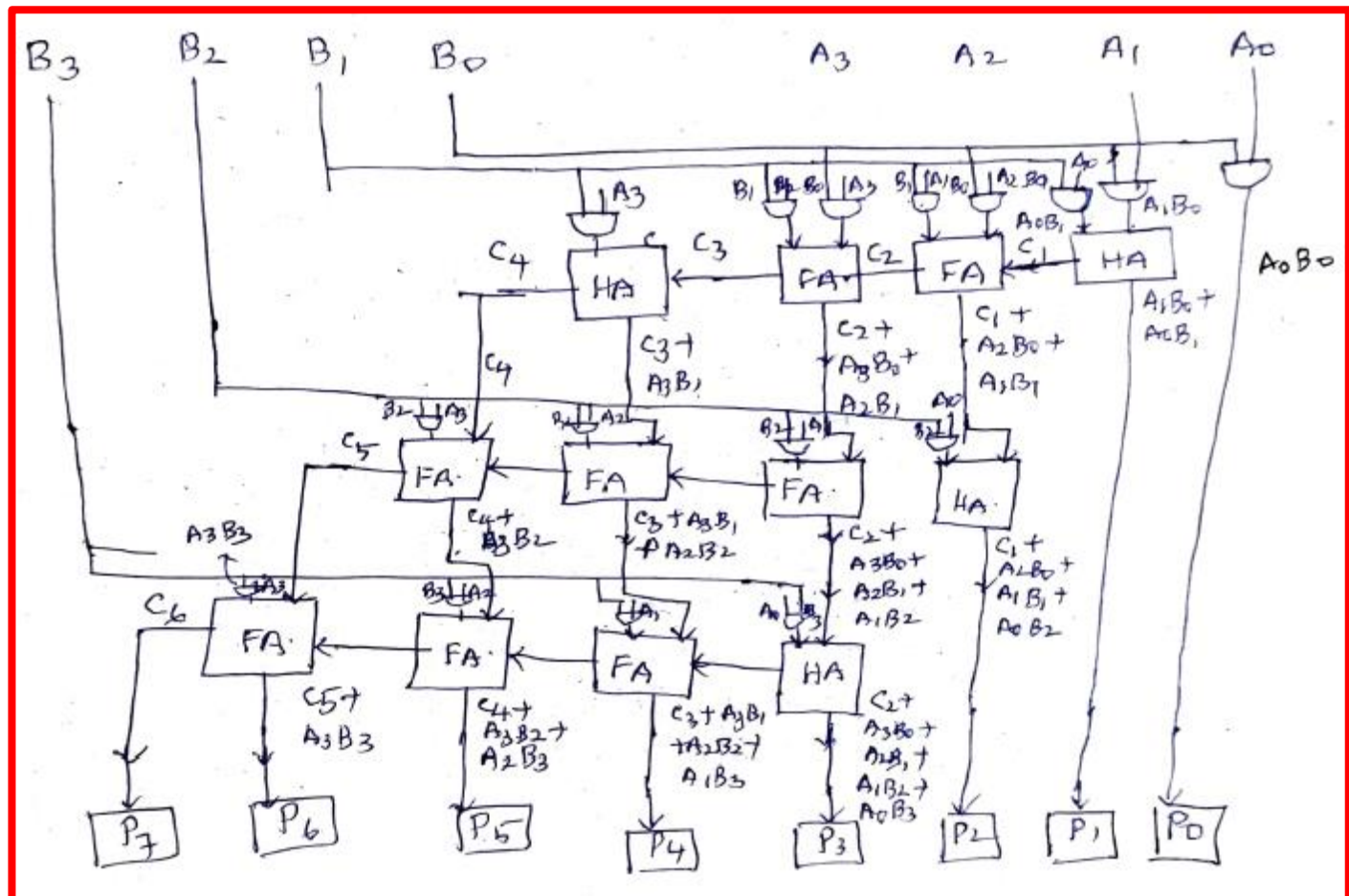
8 - FAs

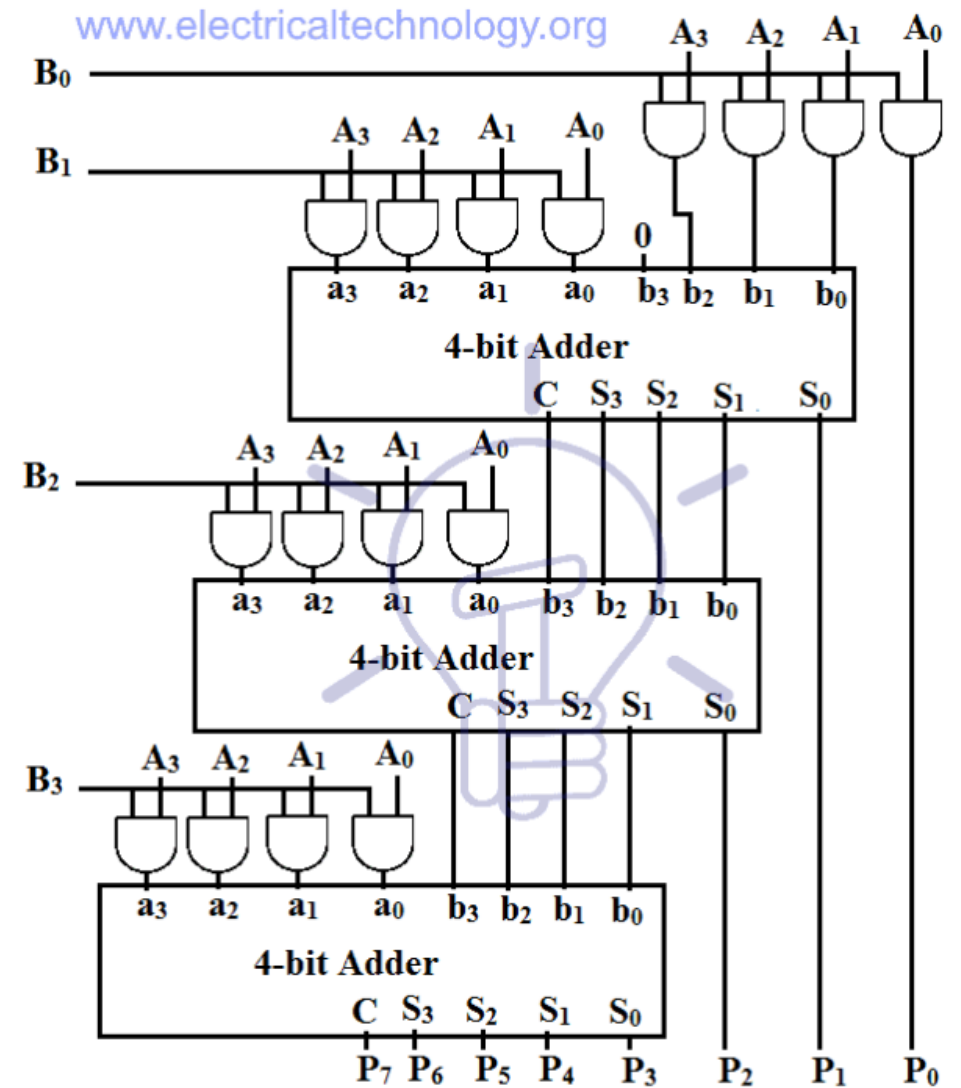
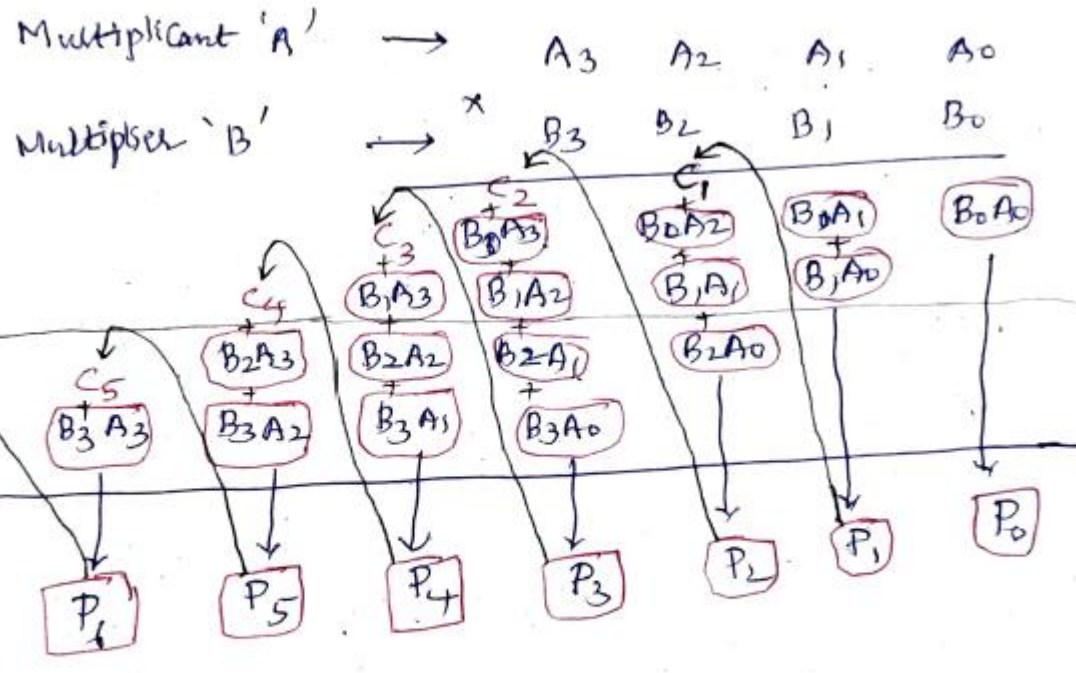
12 - adders

In general,

$m \times n$ AND gates (No. of (No. of
 n No. of HAs (No. of A bits) B bits)
 $(m-2)n$ No. of FA
 $(m-1)n$ Total No. of adders

4×4
 16 AND gates
 4 HAs
 8 FAs
 12 Adders





Schematic of 4x4 Multiplier Using 4-Bit Full Adders

Signed Multiplication, Booth's Algorithm.

Example: $(-7) \times (+3) = (-21) \rightarrow \text{Product}$
 $\downarrow$
 $M \rightarrow \text{Multiplicand}$ $\downarrow$ $\text{Multiplier} \rightarrow Q$

Timing table:

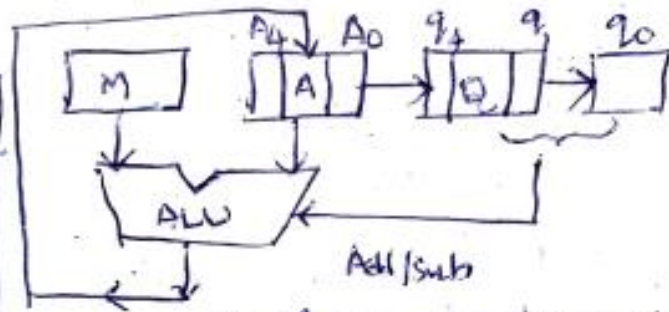
n	A	Q	q ₀	Action/Comment
4	0000	0011	0	Initialization
3	0111	0011	0	$A = A - M$
3	0011	1001	1	ASR A, Q, q ₀
2	0001	1100	1	ASR A, Q, q ₀
1	1010	1100	1	$A = A + M$
1	1101	0110	0	ASR A, Q, q ₀
0	1110	1011	0	ASR A, Q, q ₀

AQ $\rightarrow$ 1110 1011

2's of AQ
 $- 0001 0100$
 $- 0001 0101 \rightarrow (-21)$

M $\rightarrow$ 0111
 2's of $\rightarrow$ 1000
 $-7 \rightarrow$ 1001
 M $\rightarrow$ 1001

$-M \rightarrow$ 01112



Hardware Implementing Booth's algorithm

A $\leftarrow$ 0001
 1001
 1010

